

Predicting Film Financial Success Using Pre-Release Features

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Questions We Set Out to Answer

- Q1) Do genre-level historical ROI patterns carry meaningful predictive signal?
Q2) How does historical actor performance influence prediction?
Q3) Does clustering reveal interpretable actor archetypes?
Q4) Which model best classifies film success using only pre-release features?
Q5) How does model performance vary across production budget segments?
Q6) Which pre-release features have the largest impact on predictions?

Datasets & Setup

Source	Size	Period	Role
IMDb + TMDB (merged)	6,768	2000–2019	Primary
Train	4,142	2000–2012	Fit
Validation	1,142	2013–2015	Tune
Test	1,484	2016–2019	Evaluate

Target: We defined a film to be successful if Revenue $\geq 1.25 \times$ Budget (25% profit margin). This is a realistic expectation to be met based on the ROI distribution of movies in our dataset combined with a reasonable rate of return for any investment for a given timeline.

Features: We constructed 28 features combining raw pre-release metadata with engineered historical aggregates. Raw features include production budget, runtime, release year, and genre flags. Engineered features capture the historical median ROI, rating, and budget for each film's primary actors and genres, computed with matching budget buckets to reduce cross-segment noise. All historical features were built exclusively from training data to prevent leakage.

Budget Buckets: Based on the budget distribution of films in our final dataset, we decided to split films into 4 distinct buckets, keeping in mind both similarity in size of the buckets as well as similarity of films in the buckets. The four budget buckets chosen were \$0-5M, \$5-25M, \$25-100M, and \$100M+.

Imputation: Since all historical performance features were engineered using data from the train set to account for data leakage, some actors/genre combinations were never seen in the train set and therefore were unable to be computed for movies in the validation and test sets. We decided to impute with budget-bucket specific median values, rather than mean values, for these features to help account for massive outliers which artificially inflated feature values and therefore skewed predictive accuracy.

Excluded: We excluded data from 2020 onwards due to the fact that many movies were being co-released in theaters and streaming platforms. The IMDb data only captured theater revenues which caused many new movies with massive budgets to show unrealistic and incorrect true revenues.

Method Overview

- EDA:** Genre, actor, director analysis across ROI, ratings, budget over time
- Feature Engineering:** Historical medians per budget bucket; hierarchical imputation (group $\rightarrow$ global median)
- Clustering:** K-Means (Elbow + Silhouette) on actors & movies — archetype visualization only, not used as model features
- Classification:** Decision Tree · Random Forest · Logistic Regression · Gradient Boosting
- Evaluation:** ROC-AUC · Precision-Recall · Confusion Matrices · SHAP Interpretability

Limitations & Future Work

- Lack of Pre-Release Features:** No marketing spend, trailer data, theater count, festival inclusion, filming duration, amount of animation, or social media features available.
- Theater-Streaming Co-releases:** Streaming revenue often incorrect from IMDb; 2020+ excluded to mitigate.
- Class Imbalance:** \$100M+ bucket has ~85% success rate, causing the model to learn generally that high budget = success.
- Recency Bias:** Create rolling time windows to address actor recency bias, or train/test on smaller rolling windows.

Reflections

1. Difficulty in Feature Engineering: The temporal nature of our data required us to only use the train set when generating historical performance features, as these would be the only data points available to us prior to the release of a new film. If trends changed (which they did) over time, those historically aggregated features could cause inaccurate predictions for future films.

2. Model Choice/Complexity Wasn't the Issue: All four models clustered between AUC 0.62–0.66, which tells us the issue wasn't in our model choice or design, it was the lack of signal from the data and the inherent variance seen in the film industry. Reflecting on this, we'd invest more effort upfront in data enrichment rather than hyperparameter tuning.

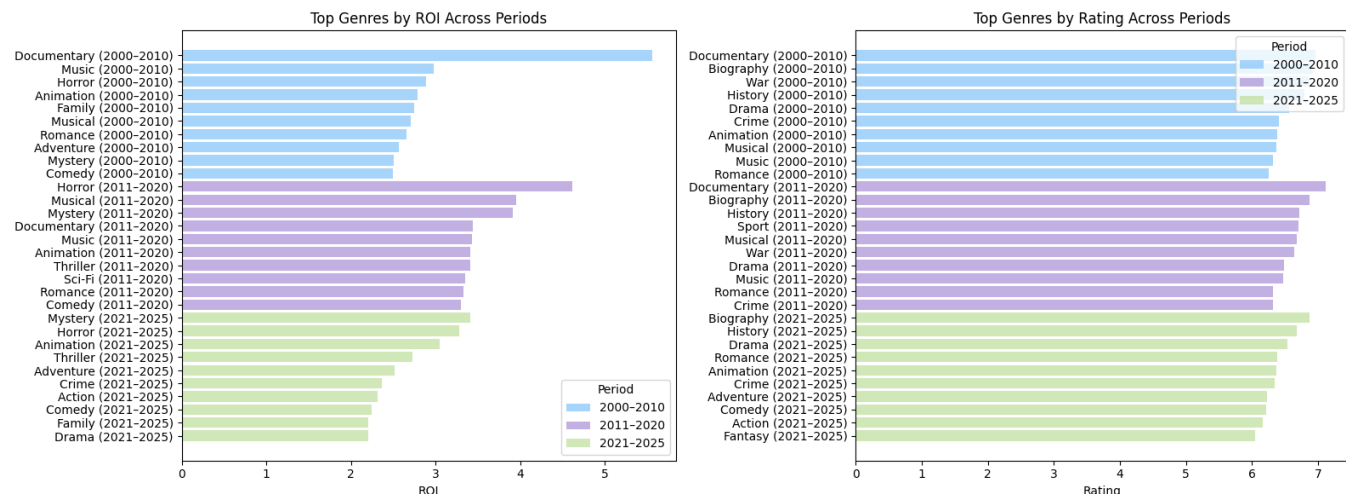
3. Changes in Trends Over Time: Over time, financial leadership shifts across genres, while audience preferences remain relatively stable. At the same time, the roles of actors and directors become increasingly specialized with clear separation between high-budget production, cost-efficient films, and critically driven projects.

4. Successes of our Approach: Our approach identifies clear structural patterns in the film industry and reveals shifts in genre performance and increasing specialization of actor and director roles over time. Using these findings as features in our model, we were able to achieve measurable improvement over random and all-success classifiers.

5. Failures of our Approach: After running our model performance metrics, specifically SHAP, we identified that our data lacked signal heavy features that would help the model make nuanced predictions within each budget bucket. For high budget buckets specifically, this caused the model to be unable to effectively differentiate between success and failure. Based on results across budget buckets, our model's improvement lift over baseline performance is not quite strong enough, yet, to be successfully used at scale.

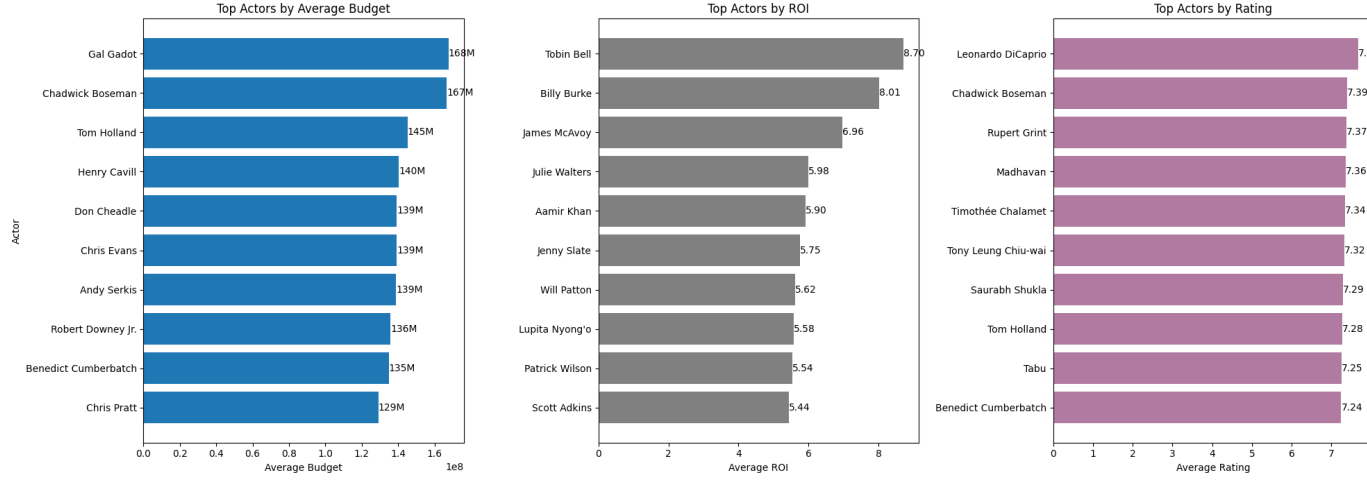
6. Who Is Affected by Our Work: The \$100M+ bucket's roughly 85% success rate caused our model to learn that high budget generally equals success, rather than identifying genuine predictive signals. Studios or investors using this model could inadvertently over-invest in high budget films while systematically underfunding viable low-budget projects.

Q1 — Genre ROI vs. Rating Divergence



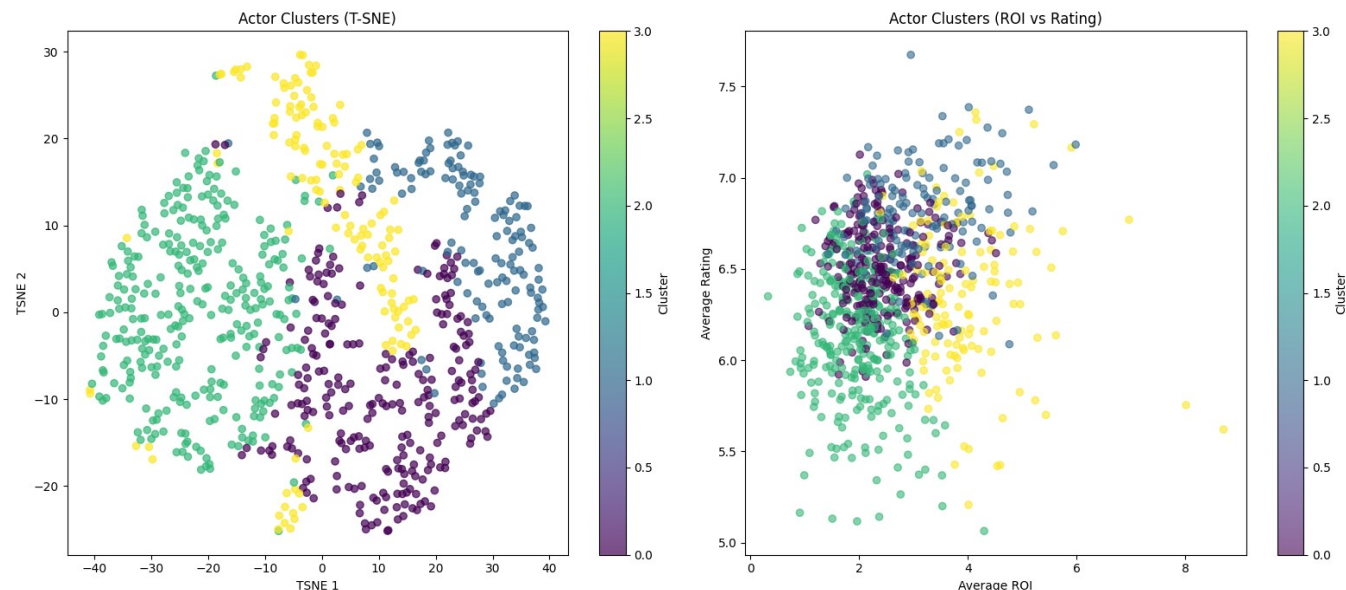
Takeaway: Documentary leads ROI in early periods; Horror/Mystery in later periods. Biography/War/History lead ratings. Financial and critical success are structurally decoupled.

Q2 — Actor Star Power Analysis



Takeaway: We have identified three main archetypes: franchise actors (high budget, e.g. Gadot, Holland), genre actors (high ROI, e.g. Tobin Bell 8.70x), and critically acclaimed (high rating, e.g. DiCaprio). However, star power alone $\neq$ financial success for a film.

Q3 — Actor Clustering Archetypes



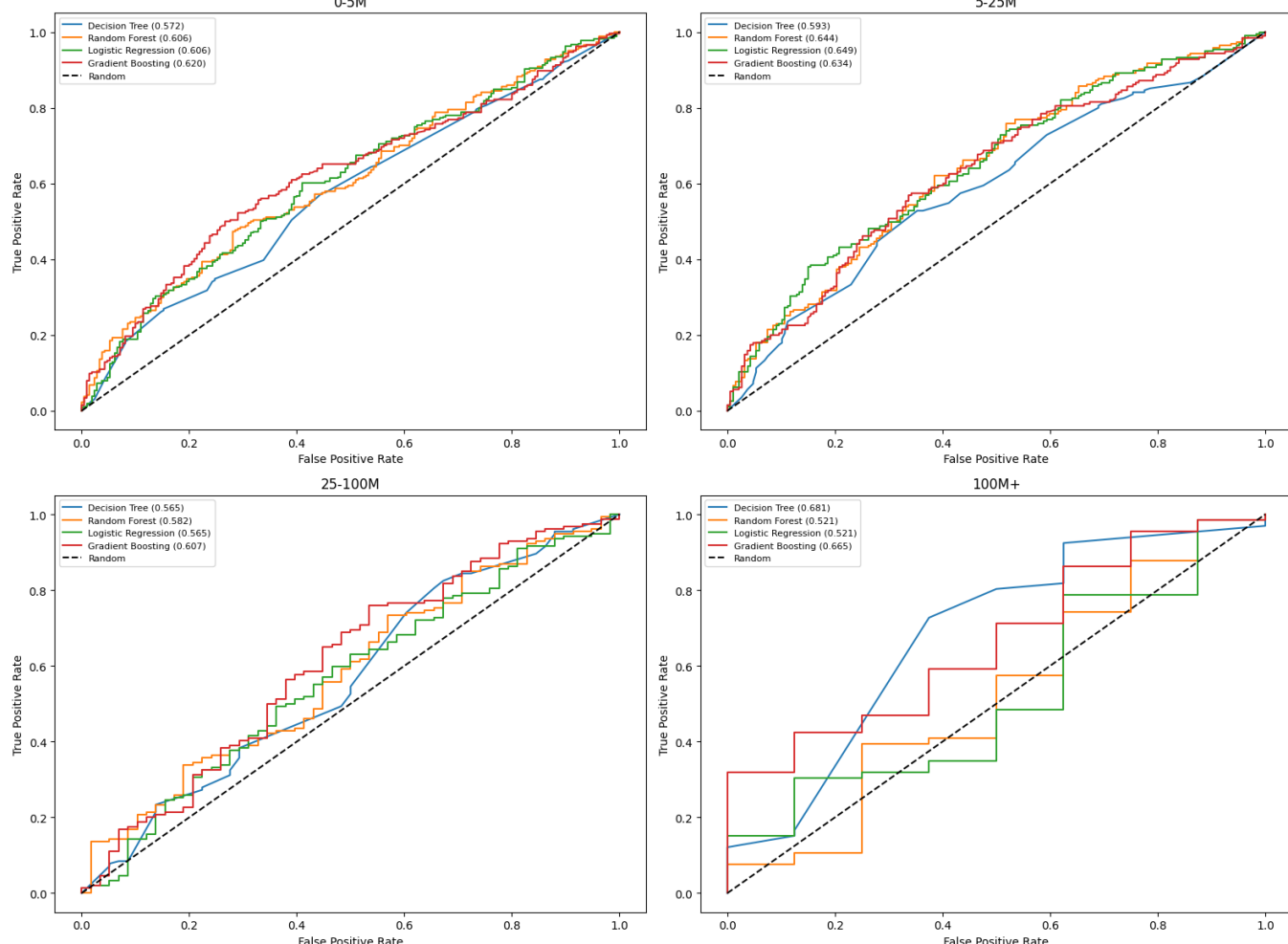
Takeaway: Clustering analysis confirms that both movies and actors form distinct archetypes rather than a single notion of success. High-budget, high-ROI, and highly rated films represent different outcome types, highlighting the multi-dimensional structure of the film industry.

Q4 — Models' Performance (Validation Set)

Model	Accuracy	Precision	Recall	F1
Gradient Boosting Classifier	0.62	0.68	0.71	0.69
Random Forest	0.61	0.68	0.65	0.67
Decision Tree	0.61	0.68	0.63	0.65
Logistic Regression	0.62	0.72	0.59	0.65

Takeaway: Gradient Boosting Classifier leads on accuracy, recall & F1, and only trails Logistic Regression in Recall due to Logistic Regression having a significant amount of false positive predictions. All models cluster around an AUC of 0.62–0.66, meaning feature signal, not architecture, is the limiting factor for our models.

ROC Curves - All Models + Budget Buckets (Validation Set)

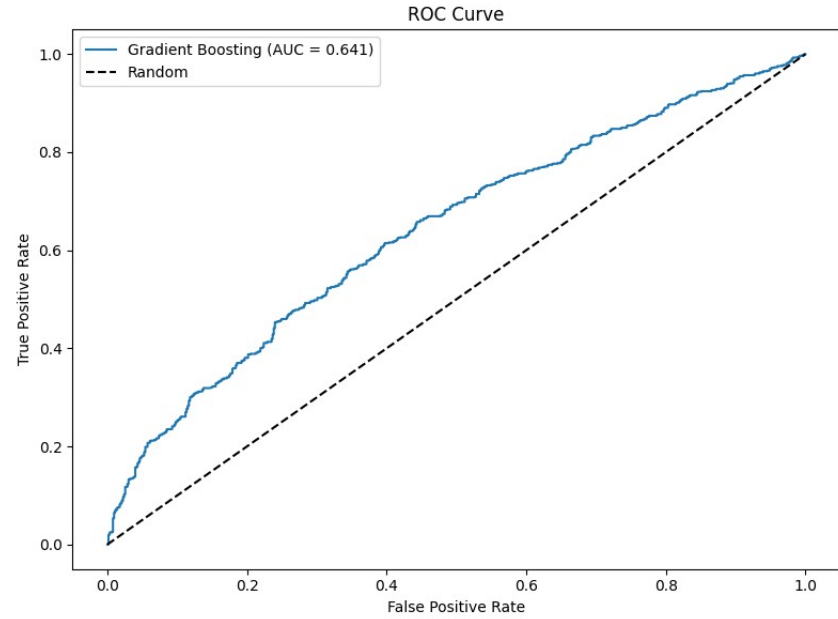


Takeaway: AUC for lower budget buckets (\$0-5M and \$5-25M) show significantly better results than the higher budget buckets. Gradient Boosting Classifier shows the best overall performance across budget buckets.

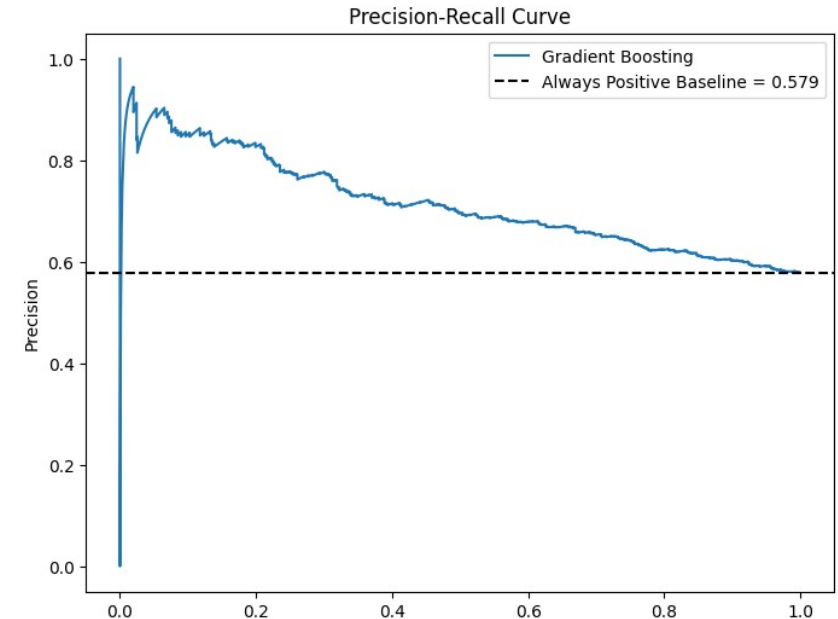
Model Results

0.641 GBC Test AUC	0.674 GBC Test F1	0.613 GBC Test Acc.	+14.1% / +7.9% vs. Random / All-Pos.
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ROC Curve — Gradient Boosting (Test Set, AUC = 0.641)



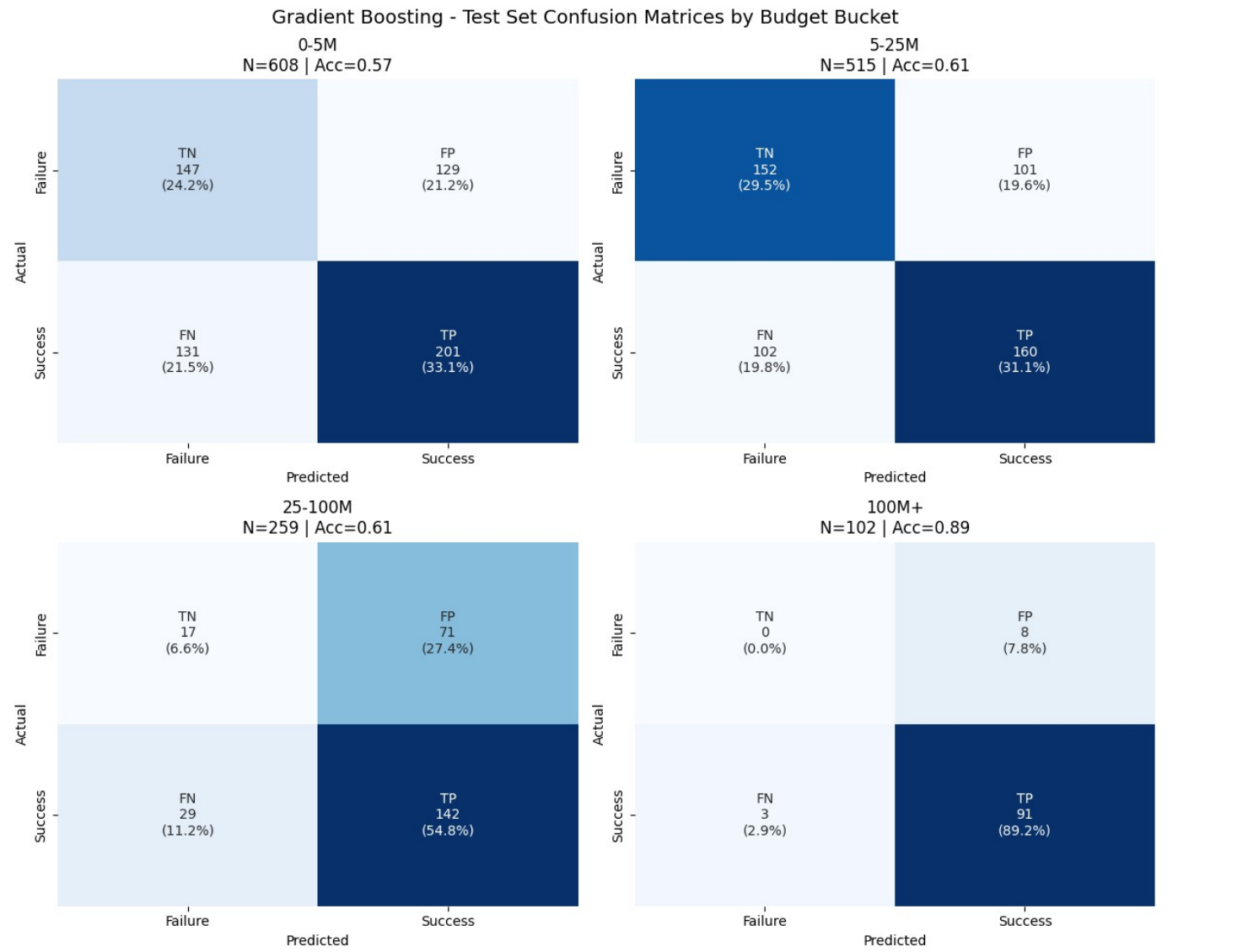
Precision-Recall Curve — Gradient Boosting (Test Set)



Takeaway: AUC shows material improvement over random prediction, and slight improvement over all-positive prediction. Precision-Recall curve shows significant drop in precision as recall increases.

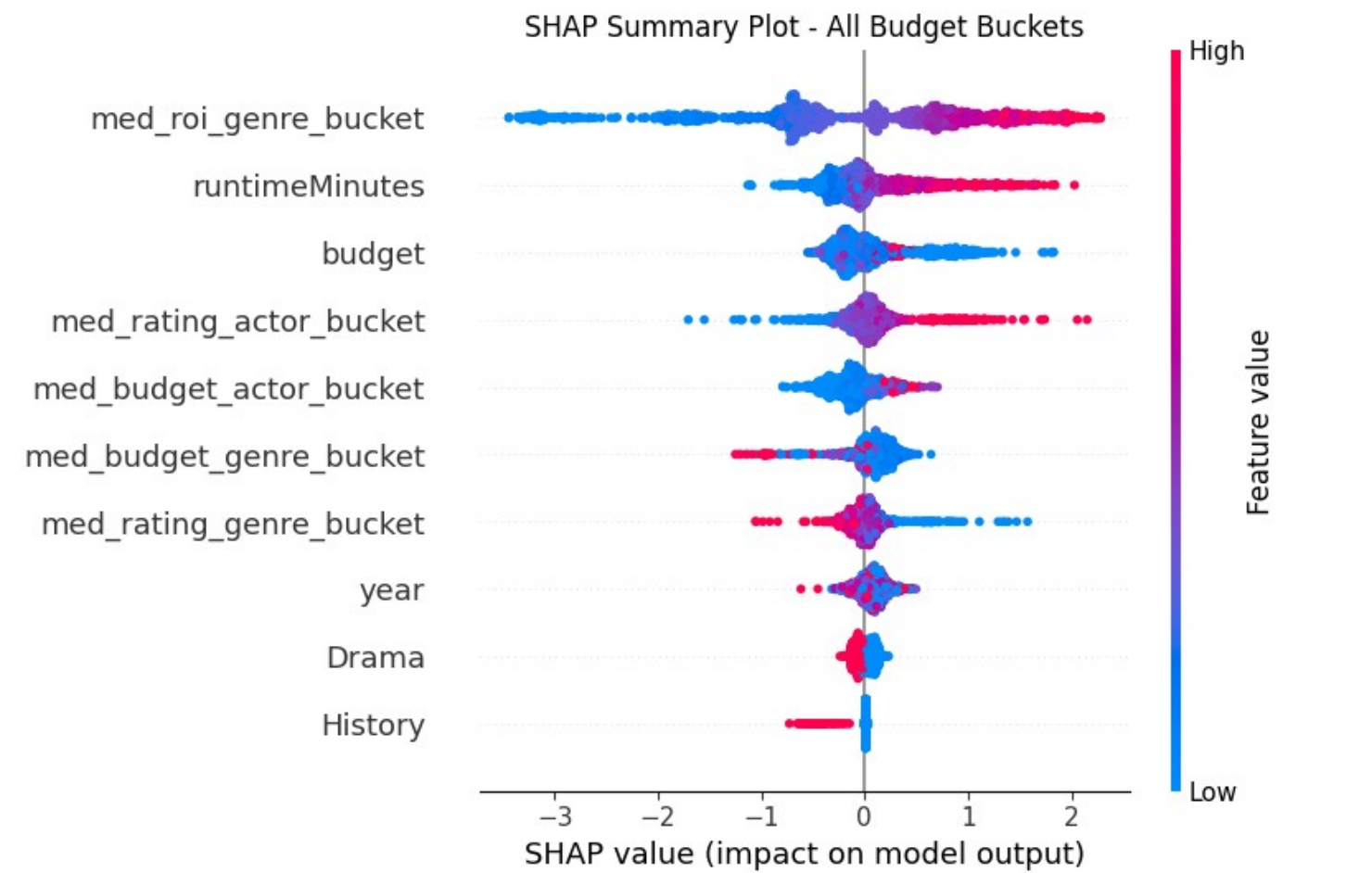
Q5 — Budget-Specific Performance (Test Set)

Budget Bucket	N	Accuracy	Precision	Recall	AUC
\$0–5M	608	0.57	0.61	0.61	0.608
\$5–25M	515	0.61	0.61	0.61	0.608
\$25–100M	259	0.61	0.67	0.83	0.547
\$100M+	102	0.89	0.92	0.97	0.535



Takeaway: \$100M+ hits 89% accuracy but AUC 0.535, meaning our model learns 'big budget = success.' Lower budget buckets show more balanced, informative predictions.

Q6 — SHAP Feature Importance, All Budget Buckets (Test Set)



Takeaway: med_roi_genre_bucket (The median ROI of all films from the training set in a respective genre and respective budget bucket) dominates all segments, showing that genre historical ROI's predictive power far outweighs raw budget and all historical actor based features.